

Note 3.7 Implicit Differentiation and Related Rates

Implicit Differentiation

- **Explicit Functions and Implicit Functions:**

- Explicit Functions: $y = f(x)$ ($y = x^2 + 1, y = ex$)
- Implicit Functions: $F(x, y) = 0$ ($x^2y^3 - 6 = 4y^2 + x$)

- **Implicit Differentiation:**

- Differentiate both sides of the equation with respect to x .
- Solve the differentiated equation algebraically for $\frac{dy}{dx}$ in terms of x and y .

- **Logarithmic Differentiation:**

The derivatives of positive functions sometimes can be more quick if we take the natural logarithm of the both sides before differentiating, the process called logarithmic Differentiation.

For

$$y = u^v$$

then

$$\begin{aligned}\ln y &= v \ln u \\ \frac{1}{y} \cdot y' &= v' \ln u + v \cdot \frac{u'}{u} \\ y' &= u^v \left(v' \ln u + v \cdot \frac{u'}{u} \right)\end{aligned}$$

Derivatives of Higher Order

- **Strategies:**

Use the approach of implicit differentiation to find the y' and then substitution the y' to the higher order derivatives.

Related Rates

- **Definition:**

There are two functions $x = x(t)$ and $y = y(t)$ related by $F(x, y) = 0$. Differentiate both sides of the equation with respect to t , we will get it involving $\frac{dy}{dt}$ and $\frac{dx}{dt}$ (the rate of change). This equation is called related rates equations

- **Strategy:**

- Draw picture and name variables and constants
- Write down the numerical information

- Write down what to find
- Write down an equation related to variables
- Differentiate with respect to t
- Evaluate.